INCH-POUND

MIL-L-24942A

9 AUGUST 1993
SUPERSEDING
MIL-L-24942
27 September 1988

MILITARY SPECIFICATION

LINER, FIREMAN'S, TROUSER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of removable, quilted heat protective liner for firemen's trousers.

1.2 Classification. The trouser liner shall be of one type only and in the following sizes and lengths (see 6.2):

Sizes: 34, 36, 38, 40, 42, 44

Lengths: Regular and Long

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (see 6.2)

SPECIFICATIONS

FEDERAL

A-A-50195	-Thread, Aramid
CCC-C-419	-Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered
DDD-L-20	- Label, For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-C-43774	- Cloth, Plain or Pajama Check Weave, Aramid
MIL-T-44100	- Thread, Para-Aramid, Spun, Intermediate Modulus
MIL-B-87002	- Batting, Synthetic Fibers, Quilted, Aramid

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection Attributes
MIL-STD-129	- Making for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-2073-1	- DOD Materials Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2	- Packaging Requirement Codes
MS27980	- Fastener, Snap, Style 2 (Regular Wire Spring Clamp Type)

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified,the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 1971 - Standard on Protective Clothing for Structural Firefighting, 1991 Edition.

NFPA 1976 - Standard on Protective Clothing for Proximity Firefighting, 1992 Edition

(Application for copies should be addressed to National Fire Protection Association, Batterymarch Park, Quincy, MA. 02269-9990.)

WARNING

THE TROUSER LINER WAS DESIGNED TO MEET NFPA STANDARDS, AND MUST MEET THIRD PARTY CERTIFICATION TO ASSURE CONSTRUCTION, COMPOSITE MATERIALS, AND PERFORMANCE REQUIREMENTS MEET SPECIFIED REQUIREMENTS. THE CONTRACTOR SHOULD OBTAIN A COPY OF APPLICABLE NFPA STANDARD FOR CERTIFICATION PROCEDURES.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

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3.3.1 Basic material. The basic material for the quilted batting shall conform to MIL-B-87002, except the needled batting filler shall be 100% para-aramid and meet the following physical requirements:

The thickness in inches at .05 p.s.i., shall range between .090 to .135. Compression recovery (percent minimum), is not applicable.

3.3.1.1 Thread, quilting. The thread used to fabricate the quilted batting shall be of a spun aramid and para-aramid staple, conforming to the following requirements below. The shade of the thread shall approximate the trouser liner material.

Top Thread (Needle)	Bottom Thread (Bobbin)
Size: 45/3 Aramid	Size: Tex 27
	Meta-Aramid
Staple Spun	Craq-Spun (2 ply)
Single End Break (2 lbs. Min.)	Single End Break (5 Lbs. Min.)
Elongation: 35% Max.	Elongation 6% Max.
Yds/lb. 10,000 Min.	Yds/lb. 20,000 Min.

The stitch pattern shall be of a chicken wire design as shown in Figure 2.

3.3.2 Thread. The thread for assembling the trouser liner shall be a 60 Tex aramid thread, natural shade, conforming to A-A-50195. As an alternate, a para-aramid thread conforming to size T-50 of MIL-T-44100, natural in color may be used.

3.3.3 Snap fasteners. The snap fasteners for attaching the trouser liner shall consist of a stud and eyelet (male half). The fasteners shall be style 2, 24 line (size 1), nickel plate or bright chrome finish, conforming to MS-27980 and the following part numbers:

Stud - 7N or 7C Eyelet - 8N or 8C

3.3.4 Facing. The waist facing for the trouser liner shall be fabricated from an unbleached cotton duck, size number 6, medium texture conforming to type I of CCC-C-419.

3.3.5 Binding. The binding for the raw edges of the trouser liner shall be bias cut from aramid cloth conforming to MIL-C-43774, and have a minimum width of 1-5/8 inches. The color shall be natural.

3.3.6 Labels.

3.3.6.1 Identification/ instruction label. Each trouser liner shall have an identification/ instruction label conforming to type VI, class 15 of DDD-L-20, except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

LINER, FIREMAN'S TROUSER, QUILTED
BASE FABRIC: ARAMID FACE
PARA-ARAMID BATTING
ARAMID BACK
CONTRACT NO. DLA-100-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor)

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CLEANING INSTRUCTIONS
Machine Wash Cold with Detergent
Tumble Dry - LOW HEAT
OR
Follow NAVEDTRA 414-01-45-81, Formula III
USE COLD WATER WASH AND LOW TUMBLE DRY TEMPERATURE

3.3.6.2 Size and stock number label. Each trouser liner shall have a size and stock number label conforming to type VI, class 2, of DDD-L-20, except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

SIZE: 38 R (EXAMPLE)
STOCK NUMBER: 8415-00-000-0000 (EXAMPLE)

NOTE: The lengths (Regular and Long) may be abbreviated as R or L.

3.3.6.3 NFPA Approval Label. Each trouser liner shall have a NFPA approved label conforming to type VI, class 3 of DDD-L-20, attached to the inside back of garment. The label shall contain the following inscription:

THIS TROUSER LINER MEETS THE REQUIREMENTS OF NFPA 1971 STANDARD ON PROTECTIVE CLOTHING FOR STRUCTURAL FIREFIGHTING, 1991 EDITION.

WARNING

FOR STRUCTURAL FIREFIGHTING OPERATIONS, THIS TROUSER LINER MUST BE WORN UNDER THE STRUCTURAL TROUSER COMPLYING WITH MIL-T-24940 FOR LIMB/TORSO PROTECTION. THIS TROUSER LINER SHALL BE NOT WORN ALONE. ATTACHMENT TO THE STRUCTURAL TROUSER IS ESSENTIAL FOR REQUIRED PROTECTION. OUTER SHELL, MOISTURE BARRIER, AND THERMAL BARRIER MEETING ALL REQUIREMENTS OF NFPA 1971 MUST BE UTILIZED AND ALL GARMENT CLOSURES MUST BE FASTENED WHEN IN USE. DO NOT KEEP THIS GARMENT IN DIRECT CONTACT WITH FLAMES. THIS GARMENT ALONE WILL NOT PROVIDE PROTECTION FOR FIRE ENTRY APPLICATIONS OR FOR PROTECTION FROM CHEMICAL, BIOLOGICAL, OR RADIOLOGICAL AGENTS. OTHER PROTECTIVE EQUIPMENT (HEAD PROTECTION, SCBA, GLOVES, FOOTWEAR, PASS) IS REQUIRED FOR PROTECTION. KEEP THIS GARMENT CLEAN AS SOILING WILL REDUCE PROTECTIVE QUALITIES.

DO NOT USE CHLORINE BLEACH

CHLORINE BLEACH WILL SIGNIFICANTLY COMPROMISE THE PROTECTION AFFORDED BY THE TEXTILE MATERIALS UTILIZED IN THE CONSTRUCTION OF THE GARMENT. USERS MUST CLEAN AND MAINTAIN ONLY IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. DO NOT STORE IN DIRECT SUNLIGHT. NO PROTECTIVE CLOTHING CAN PROVIDE COMPLETE PROTECTION FROM ALL CONDITIONS. USE EXTREME CARE FOR ALL EMERGENCY OPERATIONS. FAILURE TO COMPLY WITH THESE WARNINGS MAY RESULT IN SERIOUS INJURY OR DEATH.

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DO NOT REMOVE THIS LABEL

THIS TROUSER LINER, WHEN WORN UNDER THE STRUCTURAL TROUSER AND PROXIMITY TROUSER COVER COMPLYING WITH MIL-T-24940, MEETS NFPA 1976 STANDARD ON PROTECTIVE CLOTHING FOR PROXIMITY FIREFIGHTING - 1992 EDITION.

Classified by Underwriters Laboratories Inc. in accordance with the National Fire Protection Association Standard for Structural Fire Fighting NFPA 1971-1991.

3.4 Design. The trouser liner shall have snap fasteners along the edges of the waist and leg openings to allow insertion into the firemen's trouser. The trouser liner shall be made of quilted material which affords thermal protection. The edges of the trouser liner are bound (see Figure 1).

3.4.1 Figure. Figures 1 and 2 are furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification. The seam allowance shall be 1/2 inch,except where otherwise specified in Table I.

3.5.1 Pattern parts. The component parts of the trouser liner shall be cut from the materials specified in accordance with the following pattern parts:

Material	Pattern parts	Cut parts
Quilted batting	Back	2
	Right front	1
	Left Front	1
Cloth, duck, cotton	Waist facing	2

3.6 Construction.

3.6.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Wherever two or more methods for seam or stitch types are given for the same part of the operation, any one may be used. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the trousers. Ends of all seams produced by stitch type 401 shall be caught in all other seams or stitching. Topstitching and edgestitching shall be 1/16 to 1/8 inch from the edge. The bight of the overedge stitching shall be 3/16 inch. The guide and knife shall be set to trim only the ravelled thread ends of the fabric.

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3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backstitched or overstitched not less than 3/8 inch. The ends of a continuous line of stitching (except label) shall be overlapped not less than 1/2 inch. The ends of label stitching shall be overlapped not less than 3 stitches. Thread breaks (all stitch type) shall be repaired by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 stitch type may be repaired by using 301 stitch type, except for the seat seam which shall be resewn. When ends of overedge stitching are not caught in other seams or stitching, the ends shall be tacked.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operations requirements. The trouser liner shall be manufactured in accordance with the operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations listed provided that the finished liners are identical to those produced by following the sequence of operations as listed in Table I.

3.7.1 Size marking. The component parts of the trouser liner shall be bundled, marked, or ticketed to assure proper size assembly. The use of metal fastening devices or sewn-on tickets for marking is prohibited. Adhesive tickets, when used, shall not discolor the material and the adhesive mass shall not adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machine, rubber stamp, or pencil is allowed.

3.7.2 Abbreviations in table of operations. The abbreviations used in Table I are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double

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TABLE I - CONSTRUCTION OF TROUSER LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL BOB, LPR
1.	Cutting.				
	a. The trouser liner shall be cut in strict accordance with patterns furnished which show size, placement of component parts, directional lines and notches for proper assemble. A maximum swing of 2-1/2 inches from the directional line is permitted. Measurements shall be taken at the top and bottom of the directional line on the pattern to the selvage edge of the fabric on the even side of the lay. The difference between the two measurements shall not exceed 2-1/2 inches. Drill holes for positioning are permitted provided they are covered in the finished trouser liner.				
	b. The component parts of the trouser liner may be cut from ends, and shall approximate each other in shade within the garment.				
2.	Replacement of damaged parts.				
	During the spreading, cutting, and manufacturing operations, components having material defects, or damages that are classified as defects in section 4 shall be removed from production and replaced with nondefective and properly matched parts.				
3.	Size marking. (See 3.7.1)				
	Mark, bundle, or ticket the component parts to assure proper size assembly.				
4.	Join liner.				
	a. Stitch the outseams with notches matching, face to face	301 or 401	SSa-1	6-8	

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TABLE I - CONSTRUCTION OF TROUSER LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL BOB, LPR
4.	Join liner. (cont'd)				
	b. Overedge stitch the raw edges together.	502 or 503 or 602	SSa-1	6-8	
	c. Stitch the inseams, face to face.	301 or 401	SSa-1	6-8	
	d. Stitch the seat and crotch seams, face to face.	301 or 401	SSa-1	6-8	
	e. Overedge stitch the raw edges together.	502 or 503 602	SSa-1	6-8	
	OR			6-8	
	As an alternate, operations 4a and 4b, 4c and 4e, or 4d and 4e, may be performed in one operation by simultaneously stitching and overedging.	515 or 516 or 519	SSa-1	6-8	
5.	Attach waist facing.				
	a. Position waist facings, face to face, at center back notch, and stitch 3/8 inch from raw edges and press open.	301	SSa-1	6-8	
	b. Overedge stitch the lower raw edge and left front edge of waist facing.	502 or 503 or 602	SSa-1	6-8	
	c. Position the back side of waist facing piece against face side of the removable liner at top of quilted liner with seam at center back and stitch 3/16 from raw edge.	301	SSa-1	6-8	
	c. Topstitch the sides and lower edge of the waist facing through all plies.	301	SSa-1	6-8	

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TABLE I - CONSTRUCTION OF TROUSER LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB, LPR
6.	Bind liner.					
	a. Bind the edges of waist, fly opening and leg bottoms of the liner with appropriate binding as specified in 3.3.5. All raw edges of the binding shall be turned under and the finished dimension from the outside folded edge to turned edge shall be 3/8 (±1/16) inch. Stitching shall be 1/4 (±1/16) inch from folded edge. The raw edges of the lining assembly shall extend to the deepest fold of the binding.					
	b. Bind the waist and fly opening in one continuous operation. Begin the binding at the top of the left fly opening. To finish the binding leave approximately 1 inch of binding. Turn binding extension back and backtack.	301	BSc-1	6-8		
	c. Bind the leg opening starting near the outseam making sure the ends of the binding overlap 3/4 to 1-1/4 inch. Excess overlapped binding will be caught in the snap fastener when attached.	301	BSc-1	6-8		
7.	Attach labels.					
	a. Center the identification/ instruction label on the seat seam on the inside back (against wearer) with the top edge of label 1 to 1-1/2 inches below the waist facing. Stitch all four sides of label 1/16 to 1/8 inch from edge.	301	LSbj-1	6-8	A	A
	b. Center the size and stock number label directly below the identification/ instruction label and stitch all four sides of label 1/16 to 1/8 inch from edge. Lower edge stitching of identification/ instruction label may be used to secure the top edge of size and stock number label.	301	LSbj-1	6-8	A	A

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TABLE I - CONSTRUCTION OF TROUSER LINER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL BOB, LPR
	c. As an alternate, the two labels may be combined, provided all inscriptions are maintained.				
	d. Center NFPA approval label approximately 1-1/2 inches below size and stock number label and stitch all four sides of label 1/16 to 1/8 inch from edge.	301	LSbj-1	6-8	
8.	Attach snap fasteners.				
	a. Position and attach nine stud snap fasteners, as indicated by marks on patterns, to the waistband. Studs shall face the inside of the liner (towards the wearer).				
	b. Position and attach two stud snap fasteners to each leg bottom, one on inseam and one on outseam. Studs shall face the outside of liner. Center of stud shall be 1/2 inches from bound edge.				
9.	Clean liner.				
	a. Trim all thread ends. b. Remove all loose threads. c. Remove all spots and stains.				
	NOTE: Care shall be taken when trimming thread ends, so that damage to the fabric does not result.				

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3.8 Sizes and measurements. Sizes and measurements of the finished trouser liners shall be as shown in Table II (see Figure 1). All measurements and tolerances are expressed in inches.

Table II - Finished Measurements

SIZE	(A) WAIST	(B) SEAT	(C) INSEAM	
			REGULAR	LONG
34	36	46	26-3/4	31-3/4
36	38	48	27-3/4	32-3/4
38	40	50	27-3/4	32-3/4
40	42	52	27-3/4	32-3/4
42	44	54	27-3/4	32-3/4
44	46	56	27-3/4	32-3/4
46	48	58	28-3/4	33-3/4
Tolerance	±1	±1	± 1/2	±1/2

NOTE: The trouser liner shall be laid flat, smooth, and inside out with the top edges of the fly opening abutting.

(A) Waist - Twice the measurement taken across back at waistband facing stitching from stitch line to stitch line.

(B) Seat - Twice the measurement taken straight across the front from inside stitch line to inside stitch line 1 inch below bottom of fly.

(C) Inseam - The measurement shall be taken along the inseam from the junction of the front center seam and the seat seam to the bottom of the liner legs.

3.9 Workmanship. The finished trouser liners shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows.

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.3 through 4.4.4.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in Table III. The sample unit shall be one finished trouser liner. The lot size shall be expressed in units of trouser liners. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15.0 for Major and Minor A defects, and 40.0 for total (major, minor A&B combined) defects.

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Table III. End item visual defects

EXAMINE	DEFECT	CLASSIFICATION MAJOR	MINOR A	MINOR B
I. General	a. Any hole, cut, tear, needle chew, smash, multiple float (In any ply)	X		
	b. Darn or mend (In any ply)		X	
	c. Exposed drill hole (In any ply)			X
	d. One or more rows of stitching omitted in the quilting	X		
	e. More than 1-1/2 inches of broken or missing stitches in quilting			X
	f. Thick or thin place in batting	X		
	g. Piecing or splicing of either ply of quilting	X		
	h. Any part not cut in accordance with directional lines indicated on patterns or not in accordance with specifications	X		
	i. Three or more thread ends 1/4 inch or longer not trimmed on inside of liner		X	
	j. Five or more thread ends 1/4 inch or longer not trimmed on outside of liner			X
	k. Spot or stain on inside of liner:			
	1. Up to 1 inch in longest direction			X
	2. 1 inch or more in longest direction		X	
II. Components and Assembly	a. Any component not as specified or any defect of assembly	X		
	b. Any component, component part, or required operation omitted, or any operation improperly performed.	X		
	c. Color of component not as specified	X		
III. Seams and Stitching	a. Accuracy of seaming:			
	1. Any seam, hem or attachment of any part puckered, pleated, or caught in an unrelated operation or stitching		X	
	2. Any seam allowance not as specified			X
	3. Any end of seam and stitching when not caught in another seam or stitching:			
	(a) Not backstitched		X	
	(b) Backstitched or overstitched less than 3/8 inch			X
	4. Any seam restitched damaging cloth to a degree where it may tear along the row of stitching	X		
	5. End of continuous line of stitching overlapped less than 1/2 inch			X

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Table III. End item visual defects

EXAMINE	DEFECT	CLASSIFICATION MAJOR	MINOR A	MINOR B
III. Seams and Stitching (cont'd)	6. Stitching anchoring both ends of thread break or skipped stitches less than 1/2 inch			X
	b. Open seam:			
	1. Any seam except quilt or serge stitching		X	
	2. Quilt or serge stitching - 1 inch or more			X
	Note: A seam shall be classified as open when one or more stitches are broken or two or more continuous runoff or skipped stitches occur.			
	c. Seam and stitch type:			
	1. Not specified seam or stitch type			X
	2. Looper (underthread) on outside when 401 stitch type is used		X	
	d. Stitches per inch:			
	1. Stitches more or less than specified:			
	(a) By more than two stitches	X		
	(b) By more than one stitch			X
	Note: Score only when condition exists on more than 1/2 the length of the seam			
	e. Overedge stitching:			
	1. Omitted for 2 or more inches on one or more seams			X
	2. Bight of stitching less than 3/16 or more than 1/4 inch			X
	f. Backtack:			
	1. Missing, insecure, misplaced or not serving intended purpose		X	
	2. Loose stitch tension, not specified length or type			X
	g. Stitch tension:			
	1. Loose tension resulting in loosely exposed loops of lower or top thread			X
	2. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching):			
	(a) Quilt or serge stitching			X
	(b) Liner stitching		X	
	h. Stitch margin:			
	1. Not within specified range			X
	2. Top or edge stitching sewn close to edge resulting in damage to the material:			
	(a) Affecting appearance			X
	(b) Affecting serviceability	X		

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Table III. End item visual defects

EXAMINE	DEFECT	CLASSIFICATION MAJOR	MINOR A	MINOR B
IV. Snap Fasteners	a. Not specified type, size or finish	X		
	b. Studs not positioned or aligned as indicated on pattern	X		
	c. Missing, broken or defective	X		
	d. Not properly set or attached:			
	1. Will adequately be retained on garment and will function as intended			X
	2. Will likely become detached from garment in a short time of use	X		
	Note: Snaps shall be checked for proper function and secure clinching by opening and closing several times.			
V. Binding	a. Binding not securely caught, pulls away from quilted liner		X	
	b. Binding noticeably puckered or pleated		X	
	c. Binding tight, will not lie flat		X	
	d. Binding ends overlapped other than specified			X
	e. Width of binding other than specified			X
	f. Binding not as specified		X	
VI. Label	a. Missing, incomplete, incorrect, or illegible		X	
	b. Insecure, i.e., not stitched on all four sides			X
	c. Position not as specified			X

4.4.4. Dimensional examination of the trouser liner. The trouser liners shall be examined for dimensional defects. Any dimension not meeting the specified requirement shall be classified as a defect. The lot size shall be expressed in units of trouser liners. The sample unit shall be one trouser liner. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Width and length measurements shall be made while the trouser liners are placed on a flat surface with wrinkles and creases smoothed out. No tension shall be applied when making width and length measurements. Measurements shall be made as specified in Table II.

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4.4.5 Examining of packaging requirements. An examination shall be made to determine that packaging, packing and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defect
Marking (exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted containers.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in the sample shall be examined.

4.4.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed In terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each trouser liner shall be folded in half lengthwise with the fly on the inside and folded in half again by bringing the bottom of the trouser liner up to the top. The folded trouser liner shall measure approximately 15 inches by 15 inches. Each folded trouser liner shall be banded with a commercially acceptable paper strip across the middle, with the ends secured by gummed paper tape.

5.1.2 Commercial. Trouser liners shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. Ten trouser liners, of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Each box shall be closed, waterproofed and reinforced in accordance with appendix of PPP-B-636.

5.2.2 Level B. Ten trouser liners, of size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.3 Level C. Trouser liners preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), trouser liners, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4 way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or order, unit pack, intermediate and exterior containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.5 MIL-STD-2073. When specified, (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The trouser liner is intended to be used in conjunction with the Trousers, Fireman's, Structural/Aluminized Proximity.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Sizes, and lengths required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
d. When first article is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. When palletization is required (see 5.3).
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.5).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (key word) listing.

Batting, Aramid
Suit, Firefighters, Combination
Firefighters, Structural
Firefighters, Aluminized Proximity

Custodian:
Navy - NU
Air Force - 99

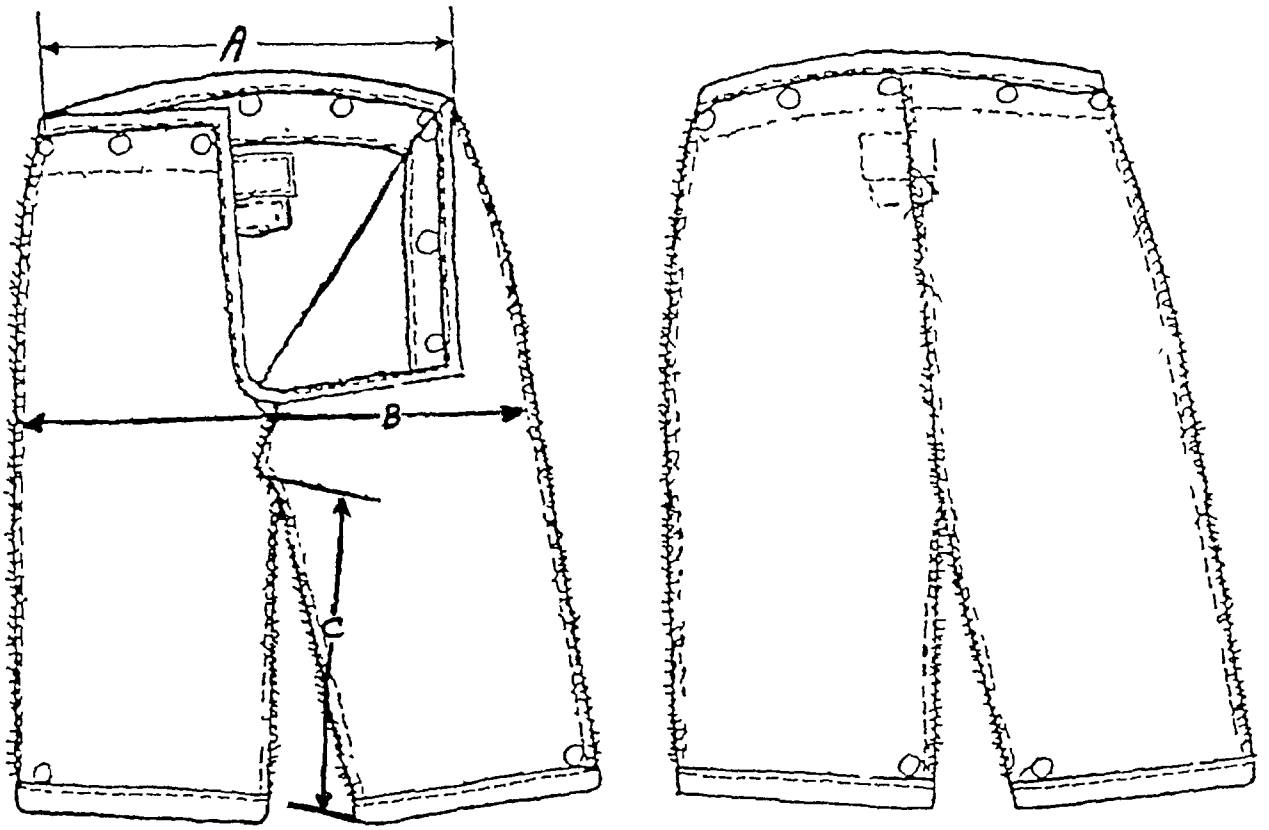
Review Activity:
Navy - MC
Air Force - 82, 50, 45

User Activity:
Navy - SH,
Coast Guard - CG

Preparing Activity:
DLA-CT

Project No.
8415-0914

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NOTE: QUILT PATTERN FOR TROUSER LINER SHOWN IN FIGURE 2

FIGURE 1
LINER, FIREMAN'S TROUSER, QUILTED

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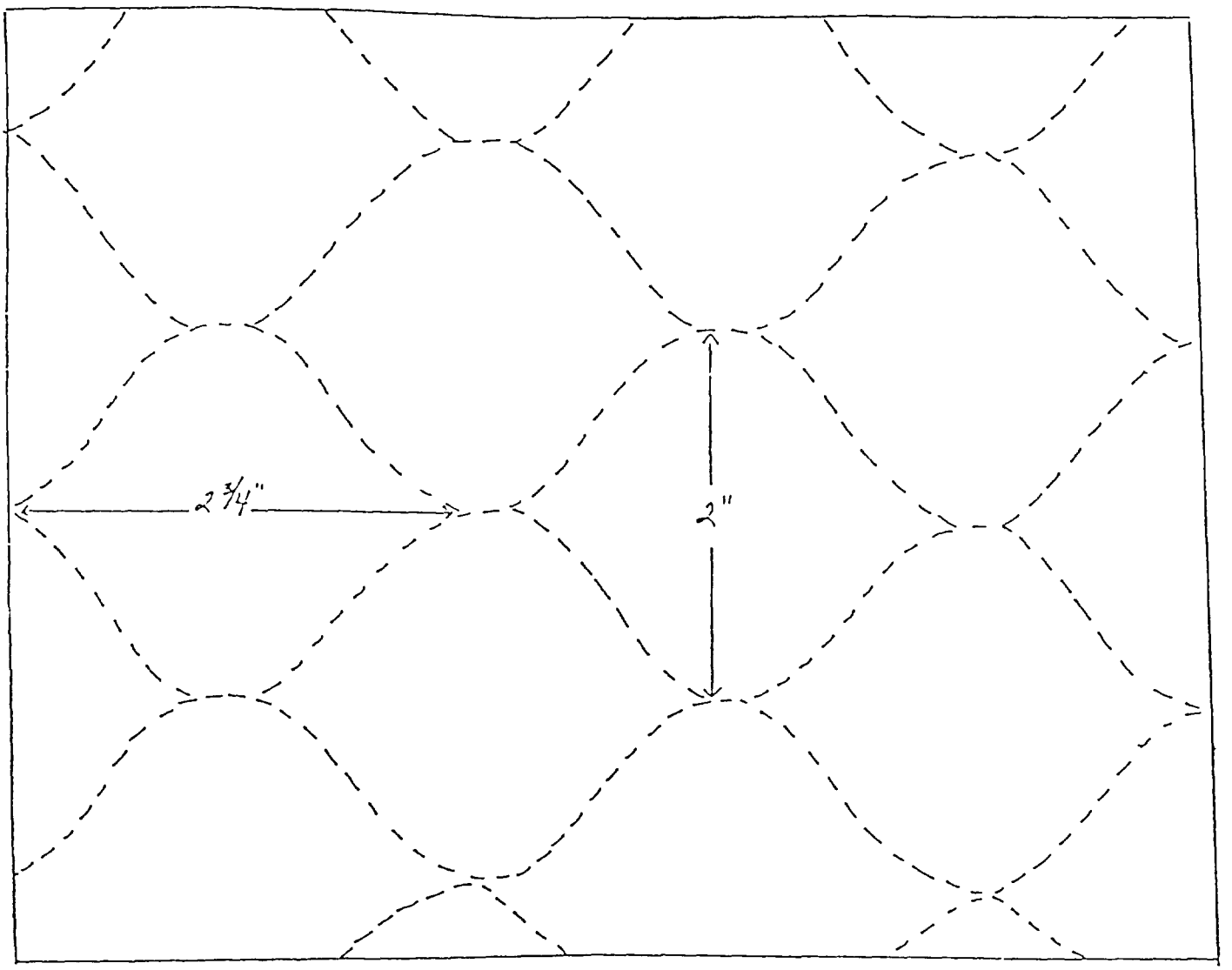


FIGURE 2
CHICKEN WIRE PATTERN

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-L-24942A	2 DOCUMENT DATE (YYMMDD) 93/08/09

3 DOCUMENT TITLE
LINER FIREMAN'S, TROUSER

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8 PREPARING ACTIVITY

a NAME Defense Personnel Support Center ATTN: DPSC-FSSD (12-3-D)	b TELEPHONE *(Include Area Code)* (1) Commercial (215)737-8105	(2) AUTOVON DSN-444-8105
c ADDRESS *(Include Zip Code)* 2800 South 20th Street P. O. Box 8419 Philadelphia, Pennsylvania 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89

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